Quadratic Kernel

Example-1

$[a^T b]^2$

$a_1 \mid b_1$
$a_2 \mid b_2$

Normal Model

Calculate $a^T b$

$a_1\ a_2 \mid b_1$
$\mid b_2$

Map Input data from 2-D to 3-D

$[a^T b]^2$

$a_1 \mid b_1$
$a_2 \mid b_2$

Vectors Mapping
a , b ⇒ u , v

Higher Computational Complexity !!!

$$\begin{bmatrix} a_1^2 \\ a_2^2 \\ \sqrt{2}\, a_1 a_2 \end{bmatrix} , \begin{bmatrix} b_1^2 \\ b_2^2 \\ \sqrt{2}\, b_1 b_2 \end{bmatrix}$$

Kernelized Model

Calculate $u^T v$

$$a_1^2\ a_2^2\ \sqrt{2}\, a_1 a_2 \mid \begin{bmatrix} b_1^2 \\ b_2^2 \\ \sqrt{2}\, b_1 b_2 \end{bmatrix}$$

Higher Computational Complexity !!!